MIL-E-1/1320B(NAVY)
11 March 1966
SUPERSEDING
MIL-E-1/1320A(NAVY)
18 December 1963
(See Note 16)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 4NP7

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Rectangular face, electrostatic focus and deflection, with integral magnetic shield

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ecl	Ebl	Eb2	ed	Rg	Zd	Alt.
Unit:	V	Vdc	Vdc	Vdc	v	Meg	Meg	ft.
Maximum:	6.9	0	1500	3000	550	1.5 (Note 1)	1.0 (Note 2)	70,000
Minimum:	5.7	-200	---	1000	---	---	---	---
TEST CONDITIONS:	6.3	Adjust	Focus	2500	---	---	1.0	---

REQUIREMENT OR TEST:

GENERAL

3.2 Qualification - Required (Note 3)

E-50.2 Holding period (t = 48 hrs. min.)

E-50.3.1 Preheating - t = 60 seconds

Section 5 Preparation for delivery

4NP7

FSC 5960

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METHOD OR PARA.	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Qualification (Note 4)					
1331	Direct interelectrode capacitance	g1 to all	C	—	10.0	uuf
		K to all	C	—	5.5	uuf
		D1 to D2	C	—	4.0	uuf
		D3 to D4	C	—	4.0	uuf
		D1 to all	C	—	10.0	uuf
		D2 to all	C	—	10.0	uuf
		D3 to all	C	—	10.0	uuf
		D4 to all	C	—	10.0	uuf
5216	Cathode illumination		—	—	—	—
5248	Deflection-factor uniformity		—	—	3	%
----	Magnetic shield	(Note 5)	—	—	—	—
1027	Temperature cycling		—	5	—	cycles
1026	Low-temperature operation	-65°C	—	—	—	—
1006	Salt spray	Tube with shield	—	—	—	—
1002	Barometric pressure, reduced	33 mmHg	—	—	—	—
3.5.2	Base material		—	—	—	—
----	Operation test	(Note 15)	—	—	—	—
	Quality conformance inspection, part 1 (Note 6)					
5106	Blemishes		—	—	—	—
5241	Grid cutoff voltage		Eco	-60	-100	Vdc
5251	Grid No. 1 leakage		—	—	—	—
----	Pattern distortion	(Note 7)	—	—	—	—
5246	Focusing voltage	cutoff	Eb1	—	710	Vdc
5246	Focusing voltage	0 Bias	Eb1	510	—	Vdc
5248	Deflection factor	1D2	DF	117	145	Vdc/in
5248	Deflection factor	3D4	DF	82	100	Vdc/in

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METHOD OR PARA.	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Quality conformance inspection, part 1 (Note 8)					
5201	Voltage breakdown		—	—	—	—
5201	Voltage breakdown (electrostatic types)		—	—	—	—
5206	Gas "cross"	Light = 8 ftL. (Note 9)	—	—	—	—
5221	Light output	(Note 9)	Light	50	—	ftL.
5223	Modulation	Light = 8 ftL. (Note 9)	Δ Ec	—	50	Vdc
5231	Spot position (electrostatic types)		—	—	7	mm
5251	Spot displacement		—	—	5	mm
5251	Anode No. 1 leakage		—	—	—	—
5251	Anode No. 2 leakage		—	—	—	—
——	Bulb to trace alinement	3D4 and major axis of face plate (Note 10)	—	—	—	—
	Quality conformance inspection, part 2					
Appendix D, 20(b)	Dimensions	See figure 1	—	—	—	—
1101	Base, cap, and insert secureness	(Note 11)	—	—	—	—
1301	Heater current		If	540	660	mA
5201	Electrode current (Anode No. 1)	Ec1 = 0	Ib1	-10	+15	μAdc
5101	Base alinement	1D2, pin 8	—	—	—	—

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METHOD OR PARA.	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNIT
	Quality conformance inspection, part 2 (Cont'd.)					
5101	Angle between traces		---	89	91	Deg
5216	Stray emission	Eb2 = 3000 Vdc	---	---	---	---
5226	Line width A	Light = 8 ftL.	width	---	0.70	mm
5226	Line width B	Light = 8 ftL (Note 12)	width	---	1.0	mm
----	Useful scan		---	3½ x 1¼	---	in.
	Quality conformance inspection, part 3					
5111	Vibration	(Note 13)	width	---	1.0	mm
1141	Pressure		---	---	---	---
1026	High temperature operation	+100°C; t = 1 hr. (Note 14)				
4.7	Life test	Group D; t = 500 hrs. (min); Eb2 = 3000V; light = 8 ftL.	---	---	---	---
4.7.3 and 4.7.3.1	Life test end points	Light = 6 ftL. (Note 9)				
		Light output	Light	40	---	ftL.
		Line width A	width	---	0.7	mm
		Line width B	width	---	1.0	mm
		Modulation	ΔEc	---	50	Vdc

NOTES:

1. Applications requiring high grid No. 1 impedance may use up to 5 megohms under controlled conditions.

2. It is recommended that the deflecting electrode circuit resistances be approximately equal. Higher resistance values up to 5 megohms may be used for low beam current operation.

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NOTES (Cont'd):

3. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

4. These tests are performed only during qualification inspection; however, qualification inspection shall consist of all of the tests listed on this specification sheet.

5. The magnetic shield shall be secured to the tube the full length, as shown on figure 1, by a non-nutrient and electrically non-conducting encapsulating material. Shielding material shall prevent spot deflection caused by a 10 gauss field from exceeding 0.5 mm with normal test voltages applied. Shield finish shall withstand a 50 hour salt-spray test in accordance with Method 101 of MIL-STD-202.

6. These shall be 100 percent inspection tests.

7. The test shall be made by means of a transparent overlay (such as photographic film) on which are two concentric rectangles 3.52 by 1.040 and 3.48 by 0.960 inches with two cross lines bisecting the rectangles and mutually perpendicular to their sides. The overlay shall be fixed in position with respect to the tube face with the center of the rectangle at the center of the tube face and with a trace produced by the 1D2 deflection plates alined with the cross line parallel with the 1D2 scan dimension of the overlay, and equal to it in length. With balanced dc voltages on the 3D4 deflection plates, the trace shall be moved out alternately to both edges of the overlay pattern. At some given trace position its entire length shall fall within the boundaries of the two rectangles. After the trace has been re-positioned to the center of the tube, and without re-positioning the overlay, a trace shall then be produced by the 3D4 deflection plates and the above procedure repeated. If the trace fails to fall within the area between the inner and outer rectangles on any of the four sides of the overlay, the tube shall be rejected.

8. The AQL for the combined defectives for attributes in quality conformance inspection, part 1, excluding inoperatives, mechanical and blemishes shall be one percent at Level II.

9. A 35- and 105-line pattern shall be provided on the cathode ray tube screen, using 60 cycles saw tooth scanning on the other axis. The mean potential of the deflection electrodes shall be that of the accelerator. The pattern size shall be adjusted to 3½ by 1 inch; the high frequency scanning shall be in the long direction of the rectangular pattern. The brightness of the area shall be measured with an eye corrected photometer, using a Wratten No. 15 filter, or equivalent.

10. The bulb and trace shall aline within 1.5 degrees.

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NOTES (Cont'd):

11. Immersion is not required.

12. Measurements shall be made at points +1.5 inches horizontal and +0.5 inch vertical from the tube face center.

13. Tube shall be rigidly mounted at base and on rectangular portion of the shield.

14. The heater shall be energized during that portion of each cycle beginning at low temperature equilibrium and ending at high temperature equilibrium. MIL-STD-202, Method 107, condition B shall be used except temperature in step 3 shall be 100° ± 3°C.

15. As a part of qualification, two tubes shall be operated in an AN/APQ-72 system for a period of six hours, and shall demonstrate mechanical and electrical compatibility and satisfactory performance in the equipment.

16. CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Review activities:

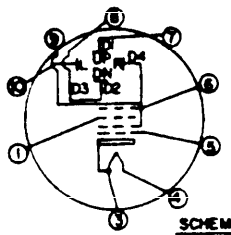
Navy - SH

User activities:

Navy - WP, MC, CG

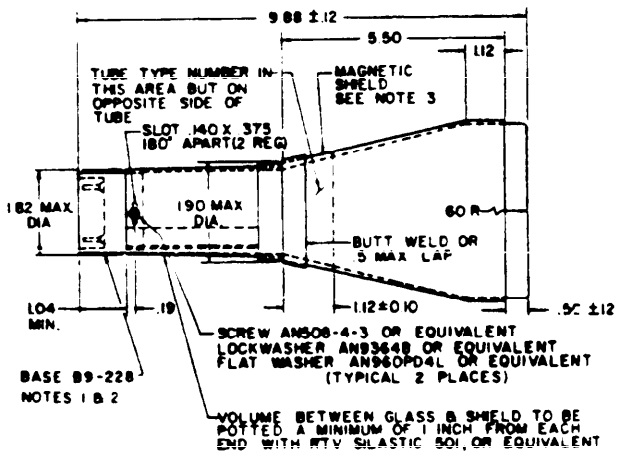
Preparing activity:
Navy - SH
(Project 5960-N648(NAVY))

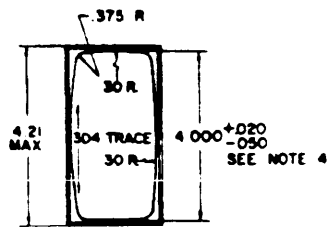
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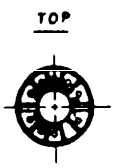


SCHEMATIC DIAGRAM
VIEWED AT FACE PLATE

PIN NO.	ELEMENT
1	ANODE NO. 1
2	-
3	HEATER AND CATHODE
4	HEATER
5	GRID NO. 1
6	ANODE NO. 2 AND GRID NO. 2
7	DEFLECTOR NO. 1
8	DEFLECTOR NO. 2
9	DEFLECTOR NO. 4
10	DEFLECTOR NO. 3







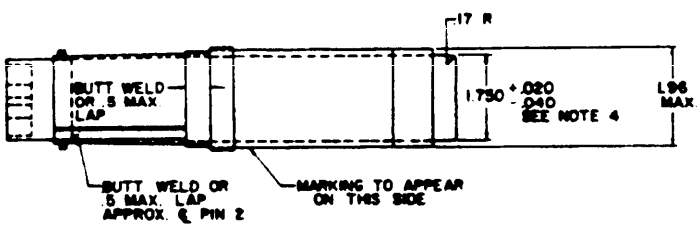


Figure 1 - Outline and dimensions of tube type [illegible]

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NOTES:

1. Insulating material: Melamine in accordance with MIL-M-14, type MME.
2. Pins: Nickel alloy, nickel plate.
3. Shield finish: Prime as required and finish as specified in MIL-E-5557, type III or IV (black).
4. Tolerance on face plate dimensions is measured on a plane 0.250 from face plate and normal to tube axis.
5. All dimensions are in inches unless otherwise specified herein.

Figure 1 - Outline and dimensions of tube type 4NP7.

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